

Advanced Optimization

Chapter 19 — Interior-Point Methods for Nonlinear Programming

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Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 19

Two lectures. Part I (§19.1–19.4): the method. Part II (§19.5–19.9): trust regions, failures, theory — and **IPOPT dissected**.

Where the whole semester was going

$$\min_x f(x) \quad \text{s.t. } c_{\mathcal{E}}(x) = 0, \quad c_{\mathcal{I}}(x) \geq 0 \quad \xrightarrow{\text{slacks}} \quad c_{\mathcal{I}}(x) - s = 0, \quad s \geq 0. \quad (19.1)$$

The hard part has never been the calculus — it is the **combinatorics** of which inequalities are active. Ch. 16/18 attacked the combinatorics head-on (active sets). Interior-point methods **dissolve it**: perturb complementarity, keep $s, z > 0$ strictly, and let the active set reveal itself in the limit.

The two-week plan

Week 14 (Part I). Two derivations of one iteration (§19.1); the basic primal-dual algorithm and its convergence anchor (§19.2); the engineering — linear algebra, μ updates, inertia correction, quasi-Newton, merit/filter (§19.3); the line search method, assembled (§19.4).

Week 15 (Part II). The trust-region method (KNITRO/CG) (§19.5); the classical primal barrier and its true failure mode (§19.6); the Wächter–Biegler counterexample and what survives it (§19.7); superlinear tuning (§19.8); software (§19.9) — ending with **IPOPT dissected**: every component of the solver the lab runs daily, mapped to a page of this book.

Lineage: Ch. 14 built this machine for LP, Ch. 16 for QP; today everything goes nonlinear — and Newton, merit functions, inertia, and projected CG report for duty one last time.

§19.1 Two derivations of the same iteration

1. Homotopy: perturb the KKT system

$$\nabla f(x) - A_{\mathcal{E}}^T y - A_{\mathcal{I}}^T z = 0, \quad Sz - \mu e = 0, \quad c_{\mathcal{E}}(x) = 0, \quad c_{\mathcal{I}}(x) - s = 0, \quad s, z > 0 \quad (19.2-19.3)$$

$\mu = 0$ is the KKT system, complementarity and all its combinatorics included. For $\mu > 0$ the system is *smooth*; under LICQ + strict complementarity + SOSC its solutions $(x(\mu), s(\mu), y(\mu), z(\mu))$ form the **primal-dual central path** $\rightarrow$ the solution as $\mu \downarrow 0$. Ch. 11's continuation idea, aimed at optimality itself.

2. Barrier: penalize the boundary

$$\min_{x,s} f(x) - \mu \sum_{i=1}^m \log s_i \quad \text{s.t.} \quad c_{\mathcal{E}}(x) = 0, \quad c_{\mathcal{I}}(x) - s = 0. \quad (19.4)$$

KKT differs from (19.2) in one line: $z = \mu S^{-1}e$ (19.5b) — rational, nasty near $s = 0$. Multiply by S : **exactly (19.2)**. Same equations, different souls.

Division of labor: the **homotopy view defines the step**, the **barrier view governs globalization**. Modern methods are *infeasible* — interior only in (s, z) ; start anywhere.

§19.2 The basic algorithm: Newton + fraction to the boundary

Newton on (19.2) in (x, s, y, z) :

$$\begin{bmatrix} \nabla_{xx}^2 \mathcal{L} & 0 & -A_{\mathcal{E}}^T & -A_{\mathcal{I}}^T \\ 0 & Z & 0 & S \\ A_{\mathcal{E}} & 0 & 0 & 0 \\ A_{\mathcal{I}} & -I & 0 & 0 \end{bmatrix} \begin{bmatrix} p_x \\ p_s \\ p_y \\ p_z \end{bmatrix} = - \begin{bmatrix} \nabla f - A_{\mathcal{E}}^T y - A_{\mathcal{I}}^T z \\ Sz - \mu e \\ c_{\mathcal{E}} \\ c_{\mathcal{I}} - s \end{bmatrix} \quad (19.6)$$

then step, but **never touch the wall**:

$$\alpha_s^{\max} = \max\{\alpha \in (0, 1] : s + \alpha p_s \geq (1 - \tau)s\}, \quad \tau = 0.995 \text{ (same for } z\text{)}. \quad (19.9)$$

Algorithm 19.1 (Fiacco–McCormick loops): inner loop drives the KKT error $E(x, s, y, z; \mu) \leq \mu$ (19.10); outer loop shrinks $\mu \leftarrow \sigma \mu$.

Theorem 19.1 (the anchor)

If the algorithm keeps escaping the inner loop and $\mu_k \rightarrow 0$, every limit point $\hat{x}$ is **feasible**; if LICQ holds at $\hat{x}$, it is a **KKT point**.

Proof shape (inequalities only): $E \rightarrow 0$ forces $c_k - s_k \rightarrow 0$, so $\hat{x}$ feasible. On inactive constraints, $s_i z_i \rightarrow \mu \rightarrow 0$ with $\hat{s}_i > 0$ kills z_i ; LICQ makes the surviving $\{z_i\}_{i \in \mathcal{A}}$ converge, and the limit is (12.34). Ch. 12, cashing out.

The one big *if*: “keeps escaping the inner loop”. Whether that is guaranteed is §19.7’s story — and it has a villain.

§19.3a Linear algebra: one system, three sizes, benign ill-conditioning

Symmetrize (19.6) with $\Sigma = S^{-1}Z$ (19.12–19.13), then eliminate:

$$\underbrace{(n+m+l+m)}_{(19.12)} \xrightarrow{\text{drop } p_s} \underbrace{\begin{bmatrix} \nabla_{xx}^2 \mathcal{L} & A_{\mathcal{E}}^T & A_{\mathcal{I}}^T \\ A_{\mathcal{E}} & 0 & 0 \\ A_{\mathcal{I}} & 0 & -\Sigma^{-1} \end{bmatrix}}_{(19.15)} \xrightarrow{\text{drop } p_z} \underbrace{\begin{bmatrix} \nabla_{xx}^2 \mathcal{L} + A_{\mathcal{I}}^T \Sigma A_{\mathcal{I}} & A_{\mathcal{E}}^T \\ A_{\mathcal{E}} & 0 \end{bmatrix}}_{(19.16) \text{ condensed}}$$

Bounds-only inequalities: $A_{\mathcal{I}}^T \Sigma A_{\mathcal{I}}$ is *diagonal* — the condensed system costs a Ch. 16 KKT solve. LBL^T (with inertia read-out) is the workhorse.

- ▶ **Primal vs. primal-dual:** Newton on (19.5) instead gives $\Sigma = \mu S^{-2}$ (19.14) — the specific guess $z = \mu S^{-1}e$. Bounded derivatives and better steps favor the primal-dual $\Sigma = S^{-1}Z$; the primal choice resurfaces in §19.6 as the classical method's soul.
- ▶ **Ill-conditioning is real but structured:** Σ 's entries fly to 0 and ∞ as $\mu \rightarrow 0$, yet stable *direct* factorizations deliver accurate steps — errors bite only near-degenerate s, z or rank-deficient matrices.
- ▶ **Iterative solves** need the scaling first: $\tilde{p}_s = S^{-1}p_s$ turns Σ into $S\Sigma S \approx \mu I$ — clustered. Then GMRES/QMR on the indefinite forms, or better: **projected CG behind the constraint preconditioner (19.17)** — Ch. 16's Algorithm 16.2, fourth deployment.

§19.3b The barrier dial, and the inertia dial

Updating μ

Fiacco–McCormick (Alg 19.1): hold μ until $E \leq \mu$, then $\mu \leftarrow \sigma\mu$, $\sigma \in \{0.2, 0.1\}$ — simple; sensitive to scaling.

Adaptive: $\mu_{k+1} = \sigma_k s_k^T z_k / m$ (19.19), σ_k from LOQO's centrality rule (19.20) or a **Mehrotra-style probe** (19.21–19.22): try the $\mu = 0$ affine step, measure its complementarity, cube the ratio — Ch. 14/16's predictor, third tour.

Nonconvexity: the inertia test (the refrain's final form)

The step is a descent direction iff $\text{diag}(\nabla_{xx}^2 \mathcal{L}, \Sigma)$ is PD on the constraint null space $\iff$ the (19.12) matrix has inertia

$$(n + m, \quad l + m, \quad 0). \quad (19.24)$$

Read the inertia off the LBL^T factors (Thm 16.3!); if wrong, refactor with

$$\nabla_{xx}^2 \mathcal{L} + \delta I, \quad \delta \text{ increased until (19.24) holds} \quad (19.25)$$

(+ a tiny $-\gamma I$ block against rank-deficient $A_{\mathcal{E}}$). Ch. 3/4/10–11/17's regularizations — **one object, many names, final appearance: literally the loop inside IPOPT**.

QN: damped BFGS/SR1 on $\nabla_{xx}^2 \mathcal{L}$ — **approximate the Lagrangian, never the barrier**; limited-memory $B = \xi I + WMW^T$ (19.29) + SMW solves. Feasible mode: reset $s \leftarrow c_{\mathcal{I}}(x)$ (19.30).

§19.4 The line search interior-point method (Algorithm 19.2)

Judge: the barrier merit function

$$\phi_\nu(x, s) = f(x) - \mu \sum_i \log s_i + \nu \|c_{\mathcal{E}}(x)\| + \nu \|c_{\mathcal{I}}(x) - s\| \quad (\text{unsquared norms; } \nu \text{ by Ch. 18's rules}), \quad (19.26)$$

or a **filter** on (barrier value, constraint violation) pairs with feasibility restoration.

Algorithm 19.2, annotated

Step	Source
direction from (19.12), inertia-corrected by (19.25)	§19.3, Thm 16.3
$\alpha_s^{\max}, \alpha_z^{\max}$ fraction-to-boundary (19.9)	the interior in “interior point”
backtrack within $(0, \alpha^{\max}]$: Armijo on ϕ_ν	Ch. 3 + Ch. 18
update (x, s, y, z) by (19.28); QN update if used	damped BFGS, Ch. 18
inner: until $E(\cdot; \mu) \leq \epsilon_\mu$; outer: $\mu \leftarrow \sigma\mu$	Fiacco–McCormick

Maratos applies here too (SOC or nonmonotone acceptance, Ch. 18). This skeleton, with a filter as judge, **is the shape of IPOPT and KNITRO/DIRECT** — but §19.7 will show it has a crack, and the patch is part of why IPOPT looks the way it does.

— end of Part I (week 14) —

§19.5 Trust-region IPM: Byrd–Omojokun moves into the barrier

Treat the barrier problem (19.4) as *equality-constrained* and run tailored TR-SQP (Ch. 18's Algorithm 18.4), with two barrier-specific twists:

The subproblem (19.31), and why it is scaled

$$\min_{p_x, p_s} \nabla f^T p_x + \frac{1}{2} p_x^T \nabla_{xx}^2 \mathcal{L} p_x - \mu e^T S^{-1} p_s + \frac{1}{2} p_s^T \Sigma p_s \quad \text{s.t. linearized constraints} = r, \quad \|(p_x, S^{-1} p_s)\|_2 \leq \Delta, \quad p_s \geq -\tau s.$$

The S^{-1} -scaled ball shrinks slack components already near zero — the region itself avoids the wall. Change variables $\tilde{p}_s = S^{-1} p_s$ (19.32–19.33): round ball, well-conditioned block $S \Sigma S \approx \mu I$. Subproblem stationarity = the primal-dual equations of Part I.

Two-phase solve, verbatim from Ch. 18

Normal: dogleg on $\|\text{linearized infeasibility}\|_2$ in 0.8Δ (+ backtrack for $v_s \geq -\frac{\tau}{2}e$); residuals define r (19.35).

Tangential: projected CG (Alg 16.2) with Steihaug exits, projections through $\hat{A} = [A_{\mathcal{E}} \ 0; \ A_{\mathcal{I}} - S]$ (19.36) — $\nabla_{xx}^2 \mathcal{L}$ is **never factored, only multiplied**. Multipliers: least-squares (19.37), pushed positive by the primal fallback $z_i \leftarrow \min(10^{-3}, \mu/s_i)$ (19.38).

Algorithm 19.4 wraps this in the ℓ_2 merit (ared/pred) + F–M outer loop. Indefinite $\nabla_{xx}^2 \mathcal{L}$ welcome, never modified. **This is KNITRO/CG** — costlier per iteration than line search, rewarded in §19.7.

§19.6 The classical primal barrier: diagnosis corrected by hindsight

Inequality-only problem: minimize $P(x; \mu) = f(x) - \mu \sum_i \log c_i(x)$ unconstrained (Framework 19.5; Frisch 1955, Fiacco–McCormick 1968), multipliers estimated by $z_i \approx \mu/c_i(x)$ (19.47).

The received wisdom, and the correction

The Hessian (19.51–19.52)

$$\nabla^2 P \approx \nabla_{xx}^2 \mathcal{L} + \frac{1}{\mu} \sum_i (z_i^*)^2 \nabla c_i \nabla c_i^T$$

blows up like the quadratic penalty's (17.19) — “ill-conditioning killed the barrier.” **Half true.** Conditioning kills SD/CG/QN — but Newton is scale-invariant, and Newton *still* struggles. The deeper problem (Example 19.1): near $x(\mu)$ the barrier is violently **non-quadratic** — straight-edged one side, circular the other. The quadratic model is simply wrong.



Fig 19.1 (schematic), $\mu = 0.01$

The repair is a déjà vu

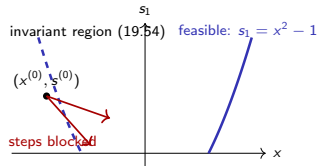
Un-eliminate: make $z_i = \mu/c_i(x)$ *variables* (19.50), add slacks — the primal barrier becomes §19.2's primal-dual method. The (17.21) move again: **when eliminating a variable poisons the nonlinearity, put it back.** Died to SQP in the 70s, reborn primal-dual in the 90s.

§19.7 The Wächter–Biegler counterexample: line search IPMs can fail

Example 19.2

$$\min x \quad \text{s.t.} \quad x^2 - s_1 = 1, \quad x - s_2 = \frac{1}{2}, \quad s_1, s_2 \geq 0.$$

Full-rank Jacobian *everywhere*. Start left of the parabola with $x^2 - s_1 \geq 1, s_1 > 0$: every step obeying the *linearized* constraints aims at the tangent line, the bound $s_1 \geq 0$ truncates it above the axis, and the region $\{x^2 - s_1 \geq 1, s_1 \geq 0\}$ is **invariant**. Iterates **never become feasible** — in practice they converge to infeasible points $(-\beta, 0, 0)$, not even stationary for infeasibility.



- ▶ Geometric failure: linearized steps + bound truncation, uncoordinated. **No merit function can fix it**; the μ schedule is irrelevant. Not Ch. 11's singular Jacobian — new.
- ▶ Remedies: when α collapses, switch to a step with a feasibility guarantee — **restoration** (filter methods) or a **trust-region step** (Alg 19.4's normal phase attacks feasibility directly; provably immune).
- ▶ The counterexample's authors are **IPOPT's authors** — filter + restoration is the institutional memory of this example.

§19.7–19.8 What can be proved, and how fast it can be

Theorem 19.2 (trust-region method, fixed μ): an honest trichotomy

With bounded $f_k, \nabla f_k, c_k, A_k, B_k$, Algorithm 19.4 reaches one of: **(i)** not asymptotically feasible — but then it converges to *stationarity of infeasibility* $\|c^-\|$ ($\nu_k \rightarrow \infty$ flags it); **(ii)** feasible but LICQ fails in the limit ($\nu_k \rightarrow \infty$ again); **(iii)** feasible, LICQ holds — KKT of the barrier problem, ν_k settles.

Combined with Thm 19.1: the full algorithm either certifies (local) infeasibility, hits a degenerate limit, or solves the problem. **Solvers' status messages are this theorem verbatim.**

§19.8 Superlinear endgame: three dials, one schedule

Near a solution with LICQ + strict complementarity + SOSC, choose

$$\mu^+ = \mu^{1+\delta} \ (\delta \in (0, 1)), \quad \tau = 1 - \mu^\beta \ (\beta > \delta), \quad \epsilon_\mu = \theta\mu \quad (19.57\text{--}19.58)$$

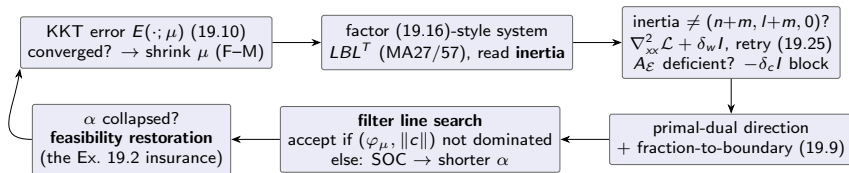
then: the full step passes fraction-to-boundary ($\alpha^{\max} = 1$), one Newton step re-satisfies the inner test, and $(x, s, y, z) \rightarrow$ solution **superlinearly**. Faster than quadratic μ -decrease buys nothing; constant σ gives only a linear rate. In practice the superlinear tail is the last few iterations — visible, brief, satisfying.

§19.9 Perspectives: the 2006 scoreboard, and what held up

Package	Type	Judge	Signature
IPOPT	line search primal-dual	filter + restoration	inertia correction; open source
KNITRO/DIRECT	line search primal-dual	merit	falls back to CG steps
KNITRO/CG	trust region (Alg 19.4)	ℓ_2 merit	projected CG; scaled TR
LOQO	line search primal-dual	merit	(19.20) centrality μ
MOSEK	convex specialist	—	conic workhorse

- ▶ **IPM vs. active-set** (the book's verdict, still current): interior-point wins large problems — every iteration solves the *same* block structure, so all engineering effort concentrates there; active-set/SQP wins warm starts and long sequences of related problems. KNITRO ships a crossover for exactly that reason.
- ▶ Augmented Lagrangian (Ch. 17): third place on general NLP, but its splitting-friendly structure is why it returned as ADMM.
- ▶ For the lab: OPF-class problems — large, sparse, cold-started, bound-heavy — sit squarely in the interior-point sweet spot. Hence the next two slides.

IPOPT dissected, I: one iteration, mapped to this book



- **Barrier loop:** monotone Fiacco–McCormick by default ($\mu \leftarrow$ superlinear schedule (19.58)-style near the end); `mu_strategy=adaptive` switches to §19.3b’s complementarity rules.
- **Step:** the condensed primal-dual system (19.16); bound-heavy problems (OPF!) make $A_{\mathcal{I}}^T \Sigma A_{\mathcal{I}}$ diagonal — the sparsity of $\nabla_{xx}^2 \mathcal{L}$ and $A_{\mathcal{E}}$ is everything.
- **Judge:** filter on (barrier objective, constraint violation) — born from Example 19.2; SOC against Maratos (Ch. 18); restoration as the last resort, itself an NLP minimizing infeasibility.
- **Options you have already met:** `hessian_approximation=limited-memory` = (19.29) = Ch. 7’s compact form; `mu_init`, `tol = E(\cdot; 0)` threshold; `warm_start_*` = fighting §19.9’s known weakness.

IPOPT dissected, II: reading the log like a native

Column	Meaning	Book object	Healthy behavior
objective	$f(x_k)$	(19.1a)	settles
inf_pr	primal infeasibility	$\ (c_{\mathcal{E}}, c_{\mathcal{I}} - s)\ $ in (19.10)	$\rightarrow 0$; stalling $\Rightarrow$ Thm 19.2(i)?
inf_du	dual infeasibility	$\ \nabla f - A^T \lambda - z\ $ in (19.10)	$\rightarrow 0$
lg(mu)	$\log_{10} \mu$	the F–M ladder (19.18)	staircase down to ~ -11
d	step norm	$\ p\ $	shrinks near the end
lg(rg)	$\log_{10} \delta_w$	inertia correction (19.25)	'-' (none); frequent $\Rightarrow$ nonconvexity biting
alpha_pr/du	accepted steps	(19.9)+filter	$\rightarrow 1$ in the endgame (19.57–58)
ls	backtracks	Armijo/filter trials	1 near the solution

- ▶ alpha_pr tagged r: iterates currently in **restoration** — Example 19.2's medicine being administered. Occasional is fine; chronic means model trouble.
- ▶ Exit messages = Theorem 19.2 verbatim: *Optimal Solution Found* (iii); *Converged to a point of local infeasibility* (i); *Solved to acceptable level* = loose-tolerance (iii).
- ▶ The endgame signature you should now expect: μ staircase accelerating, $\alpha \rightarrow 1$, $ls = 1$, superlinear drop of inf_pr/inf_du — §19.8, live.
- ▶ **Lab exercise next week:** bring one IPOPT log from your own AC-OPF runs; we read it line by line in class. Every column now has a chapter behind it.

Chapter 19 in one table

Object	Statement	Role
(19.2) $\leftrightarrow$ (19.4)	homotopy $\equiv$ barrier	step from one, globalization from the other
Alg 19.1 + Thm 19.1	F–M loops; limits are KKT	the anchor, given inner-loop escape
(19.12) $\rightarrow$ (19.16)	symmetrize, condense	Ch. 16 solves; bounds $\Rightarrow$ diagonal $A_{\mathcal{I}}^T \Sigma A_{\mathcal{I}}$
(19.24)–(19.25)	inertia test; $\delta/$ loop	nonconvexity dial; IPOPT's $\lg(\text{rg})$
Alg 19.2	line search IPM	IPOPT/LOQO/KNITRO-Direct shape
Alg 19.4	scaled-TR IPM, normal+proj. CG	KNITRO/CG; immune to Ex. 19.2
§19.6	primal barrier: nonquadratic, not just ill-cond.	un-eliminate $\Rightarrow$ primal-dual
Example 19.2	line search can never reach feasibility	birth certificate of the filter/restoration
Thm 19.2	feasible-KKT / infeasibility-stationary / LICQ-fail	solver exit messages
(19.57)–(19.58)	$\mu^{1+\delta}, \tau = 1 - \mu^\beta$	superlinear endgame

Two ideas to carry forward

Interior-point methods replace the combinatorics of active sets by analysis: one smooth system, one path, one Newton method — plus the discipline of never touching the wall. Every safeguard in a production IPM is a memorial to a specific failure: inertia correction to nonconvexity, the filter and restoration to Example 19.2, the scaled trust region to dying slacks.

Homework (1/2)

Derivations

- 1 Derive the barrier KKT system (19.5), perform the S -multiplication to reach (19.2), and explain in one sentence why Newton on the two systems gives *different* steps.
- 2 Carry out the eliminations (19.12)→(19.15)→(19.16), and verify that with bound constraints $x \geq 0$ the term $A_{\mathcal{I}}^T \Sigma A_{\mathcal{I}}$ is diagonal.
- 3 Extend the proof of Theorem 19.1 to problems with equality constraints.
- 4 Show that the condition “(19.23) PD on the null space of the constraints” is equivalent to the inertia condition (19.24), using Theorem 16.3.
- 5 Verify the invariance argument of Example 19.2: any step satisfying the linearized constraints from a point with $x^2 - s_1 \geq 1$, $s_1 > 0$, truncated by $s_1 + \alpha p_{s_1} \geq (1 - \tau)s_1$, lands again in the region (19.54).
- 6 For the schedule (19.57)–(19.58), explain why $\beta > \delta$ is needed for the full step to satisfy the fraction-to-boundary rule near the solution, and why $\mu^+ = \mu^2$ -or-faster buys nothing.

Homework (2/2)

Implementation

7. *Basic IPM.* Implement Algorithm 19.1 (direct LBL^T via `scipy ldl`, Fiacco–McCormick $\sigma = 0.2$) for inequality-constrained problems; solve Example 19.1's box problem and the circle problem (with $c \geq 0$ split). Plot the central path in the (x_1, x_2) plane for $\mu = 10^0, \dots, 10^{-6}$.
8. *Inertia in action.* Add the δI loop (19.25) and solve a nonconvex problem (e.g. $\min -x_1^2 + x_2^2$ on a box). Log δ per iteration — your own `lg(rg)` column.
9. *Break a line search IPM.* Run your item-7 code on Example 19.2 from $(x, s_1, s_2) = (-2, 3, 1)$ and watch it stall infeasibly; confirm IPOPT from the same start escapes (restoration). One experiment, one decade of algorithm design.
10. *Capstone, lab-flavored.* Solve one of your lab's AC-OPF cases with IPOPT at `print_level=5`. Annotate ten consecutive log lines: identify the μ staircase, any inertia corrections, filter rejections/SOC, and the superlinear tail. Bring it to the week-15 session — this is the exercise the whole course was pointing at.

End of the lecture track. *From Ch. 2's Taylor expansion to the solver in your terminal — nothing in that log is a black box anymore.*

Remaining self-study decks: Ch. 13–15.